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Daytime: ☒ Eveninig: ☐

(16)1- What will be the output of the following program?

```
#include <iostream>
using namespace std;
class myclass {public: double x,y;
myclass (double a, double b) {x=a;y=b;
cout<<" Constructing. x="<<x<<" y="<<y<<endl;}
myclass (const myclass &xx) {x=xx.x;y=xx.y;
cout<<"Copy Constructor. x="<<x<<" y="<<y<<endl;}
~myclass (){
cout<<"Destroying. x="<<x<<" y="<<y<<endl;}
void show() {cout<<"x="<<x<<" y="<<y<<endl;}};
void fun1(myclass ob1) {
myclass ob(3,5);ob.x=ob.x+ob1.x; ob.y=ob.y+ob1.y;
cout<<"x="<<ob.x<<" y="<<ob.y<<endl;}
int main(){
myclass ob1(8,15); ob1.show(); fun1(ob1);return 0;}
```

Constructing. 8 15

Copy Constructor.

Destroying.

(26)3- What will be the output of the following program?

```
#include <iostream>
using namespace std;
struct v1 {double x,y,z;
void set(double a,double b, double c){x=a;y=b;z=c;};
void show(){cout<<"X="<<x<<" Y="<<y<<"
Z="<<z<<endl;};
v1(double a,double b,double c) {x=a;y=b;z=c;};
v1() {x=5;y=10;z=15;};};
template <class X> void swapargs(X &a, X &b)
{X temp;temp=a; a=b; b=temp;}
void swapargs(double &a, double &b)
{cout<<"this is inside
swapargs(double,double)"<<endl;a=a*b;}
int main() {v1 xx(1,2,3), yy(10,20,30),a; int x,y;double c,d;
x=1;y=2;c=7;d=8;
cout<<"original x="<<x<<" y="<<y<<endl;
swapargs(x,y);
cout<<"swapped x="<<x<<" y="<<y<<endl;

cout<<"original c="<<c<<" d="<<d<<endl;
swapargs(c,d);
cout<<"swapped c="<<c<<" d="<<d<<endl;

xx.set(100,200,300);
cout<<"original xx: ";xx.show();
cout<<"original yy: ";yy.show();
swapargs(xx,yy);

cout<<"swapped xx: ";xx.show();
cout<<"swapped yy: ";yy.show();

cout<<"original a: ";a.show();
cout<<"original yy: ";yy.show();
swapargs(a,yy);

cout<<"swapped a: ";a.show();
cout<<"swapped yy: ";yy.show();
return 0;}
```

(15)2- Write the program code for overloading the operator "=" (assignment operator) for class class1 which is declared below. dim_a is size of the float type vector that the pointer **A points. You must take into account whether sizes of the vectors of the objects are same or not while overloading the operator '='

```
class class1 { public:int dim_a; float *A;
class1 (int a) {dim_a=a; A=new double[dim_a];}
~class1() {delete [] A;} //destructor
class1 &operator=(class1 &ob1);}
```

(16)4- What will be the output of the following program?

```
#include <iostream>
using namespace std;
class base {public: int i,j;
base (int x,int y) {i=x;j=y;}
virtual void func () {cout<<"Base version:"<<i*j<<endl;}};
class der1 :public base {
public: der1 (int x,int y) :base(x,y) {}
void func(){cout<<"using der1
version:"<<i*i+j*j<<endl;}};
class der2 :public base {
public: der2 (int x,int y) :base(x,y) {}
void func(){cout<<"using der2 version:"
cout<<i+j<<endl;}};
class der3 :public base {
public: der3 (int x,int y) :base(x,y) {} };
int main() {base *p,ob(10,20); der1 d_ob1(1,2);
der2 d_ob2(3,5); der3 d_ob3(15,30);
d_ob3.func();d_ob2.func();d_ob1.func();ob.func();
p=&ob;p->func();
p=&d_ob1;p->func();
p=&d_ob2;p->func();
p=&d_ob3;p->func();return 0;}
```

(34)6- What will be the output of the following program?

```
#include <iostream>
using namespace std;
class B1 {private : int a;public:
B1(int x) {a=x;cout<<"Base B1 a="<<a<<endl;}
B1() {a=100;cout<<"Base B1(no param) a="<<a<<endl;}
~B1() {cout<<"Destructing B1 a="<<a<<endl;}
int geta() {return a;}};
class B2 { private : int b;
public:B2(int x) {b=x;cout<<"Base B2 b="<<b<<endl;}};
~B2() {cout<<"Destructing B2 b="<<b<<endl;}
int getb(){return b;}};
class D :public B1,public B2 { private : int c;
public: D(int x,int y) :B2(x), B1()
{c=y;cout<<"Constructing D (no parameter) c="<<c<<endl;}
D(int x,int y, int z) :B2(x), B1(y)
{c=z;cout<<"Constructing D c="<<c<<endl;}};
~D() {cout<<"Destructing D c="<<c<<endl;}
void show(){
cout<<"a="<<geta()<<" , b="<<getb()<<" , c="<<c<<endl;}}.
int main () {int i;
D ob(1,2,3);
D ob1(11,22);
ob.show();
B1 bo1(50);
cout<<"a of bo1="<<bo1.geta()<<endl;
cout<<"a of bo="<<ob.geta()<<" , b of bo="<<ob.getb();
cout<<endl; return 0;}
```

(10)5- Why do we use virtual base classes? Explain.